

etana Biotech PT ETANA BIOTECHNOLOGIES INDONESIA	LOGBOOK / LOGBOOK	Document No. : SOP/EBI/EN-015-F01H
	PEMANTAUAN HARIAN TERHADAP GENERATOR AIR MURNI DAN DISTRIBUSI SKIDNYA DAILY MONITORING OF PW GENERATOR AND THE DISTRIBUTION SKID	

Date :

No	Checklist Items	Standard Value	Time : (Shift 1, Shift 2 dan Shift 3)						Remark
			07:30	11:30	15:30	19:30	23:30	03:30	
1	Piping, Pneumatic hose, Valves	No Leaks							
2	Compressed Air Pressure	0.6-0.8 MPa							
3	Availability of cooling water	Available							
4	NaClO	> 20 L							
5	NaOH	> 20 L							
6	Plant Steam Pressure (on the header)	3-4 Bar							
7	Press. Diff. of 5 Micron Filter Circulation On	≤ 7 psi							
	Press. Diff. of 5 Micron Filter Idle/Standby	Monitoring Only							
8	UV Lamp Hour Meter	< 8500 h							
9	PW Temp. On the PW Tank	≤ 40 °C							
10	PW Temp. On the looping return 1st floor	Monitoring Only							
11	PW Temp. On the looping return 2nd floor	Monitoring Only							
12	PW Conductivity after EDI Circulation On	< 1.2 µ/cm							
	PW Conductivity after EDI Idle/Stanby	Monitoring Only							
13	PW Cond. On the looping return 1st floor	< 1.2 µ/cm							
14	PW Cond. On the looping return 2nd floor	< 1.2 µ/cm							
15	PW Loop 1 Flow Velocity	> 1 m/s							
16	PW Loop 2 Flow Velocity	> 1 m/s							

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17	pH		7-9							
18	Soft Water total hardnees (once a day)		< 10 ppm							
19	Differential Pressure MMF		< 0.2 MPa							
20	Differential Pressure ACF		< 0.2 MPa							
21	Cleanliness of machine and room		Clean							
						Operator : Shift 1 Shift 2 Shift 3			Checked By :	
						() () ()			()	

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Notes :

PW Conductivity dan PW Conductivity action limit : Sesuai Lampiran SOP/EBI/EN-015-L07

UV Lamp hour meter alert limit : 8000 h, UV Lamp hour meter action limit : 8500 h

PW Loop Flow Velocity alert limit : 1.1 m/s, PW Loop Flow Velocity action limit : 1 m/s

Pressure Diff. of 5 micron filter alert limit : 6 psi

Pressure Diff. of 5 micron filter action limit : 7 psi

Pembuatan NaOH dan NaClO mengacu pada SOP/EBI/EN-015. point 7.1.1.2 dan 7.1.1.3.

Operator :

Shift 1	Shift 2	Shift 3
()	()	()

Checked By :

()

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 Pembuatan NaOH dan NaClO mengacu pada SOP/EBI/EN-015. point 7.1.1.2 dan 7.1.1.3.

Operator :

Shift 1

Shift 2

Shift 3

Checked By :

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20	Differential Pressure ACF	< 0.2 MPa							
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Notes : PW Conductivity dan PW Conductivity action limit : Sesuai Lampiran SOP/EBI/EN-015-L07 UV Lamp hour meter alert limit : 8000 h, UV Lamp hour meter action limit : 8500 h PW Loop Flow Velocity alert limit : 1.1 m/s, PW Loop Flow Velocity action limit : 1 m/s Pressure Diff. of 5 micron filter alert limit : 6 psi Pressure Diff. of 5 micron filter action limit : 7 psi Pembuatan NaOH dan NaClO mengacu pada SOP/EBI/EN-015. point 7.1.1.2 dan 7.1.1.3.			Operator : Shift 1 Shift 2 Shift 3 () () ()						Checked By : ()

 PT ETANA BIOTECHNOLOGIES INDONESIA	LOGBOOK / LOGBOOK						Document No. : SOP/EBI/EN-015-F01H
	PEMANTAUAN HARIAN TERHADAP GENERATOR AIR MURNI DAN DISTRIBUSI SKIDNYA DAILY MONITORING OF PW GENERATOR AND THE DISTRIBUTION SKID						

Date :

No	Checklist Items	Standard Value	Time : (Shift 1, Shift 2 dan Shift 3)						Remark
			07:30	11:30	15:30	19:30	23:30	03:30	
1	Piping, Pneumatic hose, Valves	No Leaks							
2	Compressed Air Pressure	0.6-0.8 MPa							
3	Availability of cooling water	Available							
4	NaClO	> 20 L							
5	NaOH	> 20 L							
6	Plant Steam Pressure (on the header)	3-4 Bar							
7	Press. Diff. of 5 Micron Filter Circulation On	≤ 7 psi							
	Press. Diff. of 5 Micron Filter Idle/Standby	Monitoring Only							
8	UV Lamp Hour Meter	< 8500 h							
9	PW Temp. On the PW Tank	≤ 40 °C							
10	PW Temp. On the looping return 1st floor	Monitoring Only							
11	PW Temp. On the looping return 2nd floor	Monitoring Only							
12	PW Conductivity after EDI Circulation On	< 1.2 µ/cm							
	PW Conductivity after EDI Idle/Stanby	Monitoring Only							
13	PW Cond. On the looping return 1st floor	< 1.2 µ/cm							
14	PW Cond. On the looping return 2nd floor	< 1.2 µ/cm							
15	PW Loop 1 Flow Velocity	> 1 m/s							
16	PW Loop 2 Flow Velocity	> 1 m/s							
17	pH	7-9							
18	Soft Water total hardnees (once a day)	< 10 ppm							
19	Differential Pressure MMF	< 0.2 MPa							
20	Differential Pressure ACF	< 0.2 MPa							
21	Cleanliness of machine and room	Clean							

Notes : PW Conductivity dan PW Conductivity action limit : Sesuai Lampiran SOP/EBI/EN-015-L07 UV Lamp hour meter alert limit : 8000 h, UV Lamp hour meter action limit : 8500 h PW Loop Flow Velocity alert limit : 1.1 m/s, PW Loop Flow Velocity action limit : 1 m/s Pressure Diff. of 5 micron filter alert limit : 6 psi Pressure Diff. of 5 micron filter action limit : 7 psi Pembuatan NaOH dan NaClO mengacu pada SOP/EBI/EN-015. point 7.1.1.2 dan 7.1.1.3.	Operator : Shift 1 Shift 2 Shift 3			Checked By : () () ()
	() () ()			

 PT ETANA BIOTECHNOLOGIES INDONESIA	LOGBOOK / LOGBOOK						Document No. : SOP/EBI/EN-015-F01H
	PEMANTAUAN HARIAN TERHADAP GENERATOR AIR MURNI DAN DISTRIBUSI SKIDNYA DAILY MONITORING OF PW GENERATOR AND THE DISTRIBUTION SKID						

Date :

No	Checklist Items	Standard Value	Time : (Shift 1, Shift 2 dan Shift 3)						Remark
			07:30	11:30	15:30	19:30	23:30	03:30	
1	Piping, Pneumatic hose, Valves	No Leaks							
2	Compressed Air Pressure	0.6-0.8 MPa							
3	Availability of cooling water	Available							
4	NaClO	> 20 L							
5	NaOH	> 20 L							
6	Plant Steam Pressure (on the header)	3-4 Bar							
7	Press. Diff. of 5 Micron Filter Circulation On	≤ 7 psi							
	Press. Diff. of 5 Micron Filter Idle/Standby	Monitoring Only							
8	UV Lamp Hour Meter	< 8500 h							
9	PW Temp. On the PW Tank	≤ 40 °C							
10	PW Temp. On the looping return 1st floor	Monitoring Only							
11	PW Temp. On the looping return 2nd floor	Monitoring Only							
12	PW Conductivity after EDI Circulation On	< 1.2 µ/cm							
	PW Conductivity after EDI Idle/Stanby	Monitoring Only							
13	PW Cond. On the looping return 1st floor	< 1.2 µ/cm							
14	PW Cond. On the looping return 2nd floor	< 1.2 µ/cm							
15	PW Loop 1 Flow Velocity	> 1 m/s							
16	PW Loop 2 Flow Velocity	> 1 m/s							
17	pH	7-9							
18	Soft Water total hardnees (once a day)	< 10 ppm							
19	Differential Pressure MMF	< 0.2 MPa							
20	Differential Pressure ACF	< 0.2 MPa							
21	Cleanliness of machine and room	Clean							
Notes : PW Conductivity dan PW Conductivity action limit : Sesuai Lampiran SOP/EBI/EN-015-L07 UV Lamp hour meter alert limit : 8000 h, UV Lamp hour meter action limit : 8500 h PW Loop Flow Velocity alert limit : 1.1 m/s, PW Loop Flow Velocity action limit : 1 m/s Pressure Diff. of 5 micron filter alert limit : 6 psi Pressure Diff. of 5 micron filter action limit : 7 psi Pembuatan NaOH dan NaClO mengacu pada SOP/EBI/EN-015. point 7.1.1.2 dan 7.1.1.3.			Operator : Shift 1 Shift 2 Shift 3 () () ()						Checked By : ()

 PT ETANA BIOTECHNOLOGIES INDONESIA	LOGBOOK / LOGBOOK						Document No. : SOP/EBI/EN-015-F01H
	PEMANTAUAN HARIAN TERHADAP GENERATOR AIR MURNI DAN DISTRIBUSI SKIDNYA DAILY MONITORING OF PW GENERATOR AND THE DISTRIBUTION SKID						

Date :

No	Checklist Items	Standard Value	Time : (Shift 1, Shift 2 dan Shift 3)						Remark
			07:30	11:30	15:30	19:30	23:30	03:30	
1	Piping, Pneumatic hose, Valves	No Leaks							
2	Compressed Air Pressure	0.6-0.8 MPa							
3	Availability of cooling water	Available							
4	NaClO	> 20 L							
5	NaOH	> 20 L							
6	Plant Steam Pressure (on the header)	3-4 Bar							
7	Press. Diff. of 5 Micron Filter Circulation On	≤ 7 psi							
	Press. Diff. of 5 Micron Filter Idle/Standby	Monitoring Only							
8	UV Lamp Hour Meter	< 8500 h							
9	PW Temp. On the PW Tank	≤ 40 °C							
10	PW Temp. On the looping return 1st floor	Monitoring Only							
11	PW Temp. On the looping return 2nd floor	Monitoring Only							
12	PW Conductivity after EDI Circulation On	< 1.2 μ/cm							
	PW Conductivity after EDI Idle/Stanby	Monitoring Only							
13	PW Cond. On the looping return 1st floor	< 1.2 μ/cm							
14	PW Cond. On the looping return 2nd floor	< 1.2 μ/cm							
15	PW Loop 1 Flow Velocity	> 1 m/s							
16	PW Loop 2 Flow Velocity	> 1 m/s							
17	pH	7-9							
18	Soft Water total hardnees (once a day)	< 10 ppm							
19	Differential Pressure MMF	< 0.2 MPa							
20	Differential Pressure ACF	< 0.2 MPa							
21	Cleanliness of machine and room	Clean							

Notes :

PW Conductivity dan PW Conductivity action limit : Sesuai Lampiran SOP/EBI/EN-015-L07

UV Lamp hour meter alert limit : 8000 h, UV Lamp hour meter action limit : 8500 h

PW Loop Flow Velocity alert limit : 1.1 m/s, PW Loop Flow Velocity action limit : 1 m/s

Pressure Diff. of 5 micron filter alert limit : 6 psi

Pressure Diff. of 5 micron filter action limit : 7 psi

Pembuatan NaOH dan NaClO mengacu pada SOP/EBI/EN-015. point 7.1.1.2 dan 7.1.1.3.

Operator :

Shift 1	Shift 2	Shift 3
()	()	()

Checked By :

()

 PT ETANA BIOTECHNOLOGIES INDONESIA	LOGBOOK / LOGBOOK		Document No. : SOP/EBI/EN-015-F01H
	PEMANTAUAN HARIAN TERHADAP GENERATOR AIR MURNI DAN DISTRIBUSI SKIDNYA DAILY MONITORING OF PW GENERATOR AND THE DISTRIBUTION SKID		

Date :

No	Checklist Items	Standard Value	Time : (Shift 1, Shift 2 dan Shift 3)						Remark
			07:30	11:30	15:30	19:30	23:30	03:30	
1	Piping, Pneumatic hose, Valves	No Leaks							
2	Compressed Air Pressure	0.6-0.8 MPa							
3	Availability of cooling water	Available							
4	NaClO	> 20 L							
5	NaOH	> 20 L							
6	Plant Steam Pressure (on the header)	3-4 Bar							
7	Press. Diff. of 5 Micron Filter Circulation On	≤ 7 psi							
	Press. Diff. of 5 Micron Filter Idle/Standby	Monitoring Only							
8	UV Lamp Hour Meter	< 8500 h							
9	PW Temp. On the PW Tank	≤ 40 °C							
10	PW Temp. On the looping return 1st floor	Monitoring Only							
11	PW Temp. On the looping return 2nd floor	Monitoring Only							
12	PW Conductivity after EDI Circulation On	< 1.2 µ/cm							
	PW Conductivity after EDI Idle/Stanby	Monitoring Only							
13	PW Cond. On the looping return 1st floor	< 1.2 µ/cm							
14	PW Cond. On the looping return 2nd floor	< 1.2 µ/cm							
15	PW Loop 1 Flow Velocity	> 1 m/s							
16	PW Loop 2 Flow Velocity	> 1 m/s							
17	pH	7-9							
18	Soft Water total hardnees (once a day)	< 10 ppm							
19	Differential Pressure MMF	< 0.2 MPa							
20	Differential Pressure ACF	< 0.2 MPa							
21	Cleanliness of machine and room	Clean							

Notes : PW Conductivity dan PW Conductivity action limit : Sesuai Lampiran SOP/EBI/EN-015-L07 UV Lamp hour meter alert limit : 8000 h, UV Lamp hour meter action limit : 8500 h PW Loop Flow Velocity alert limit : 1.1 m/s, PW Loop Flow Velocity action limit : 1 m/s Pressure Diff. of 5 micron filter alert limit : 6 psi Pressure Diff. of 5 micron filter action limit : 7 psi Pembuatan NaOH dan NaClO mengacu pada SOP/EBI/EN-015. point 7.1.1.2 dan 7.1.1.3.	Operator : Shift 1 Shift 2 Shift 3 () () ()	Checked By : ()
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 PT ETANA BIOTECHNOLOGIES INDONESIA	LOGBOOK / LOGBOOK						Document No. : SOP/EBI/EN-015-F01H
	PEMANTAUAN HARIAN TERHADAP GENERATOR AIR MURNI DAN DISTRIBUSI SKIDNYA DAILY MONITORING OF PW GENERATOR AND THE DISTRIBUTION SKID						

Date :

No	Checklist Items	Standard Value	Time : (Shift 1, Shift 2 dan Shift 3)						Remark
			07:30	11:30	15:30	19:30	23:30	03:30	
1	Piping, Pneumatic hose, Valves	No Leaks							
2	Compressed Air Pressure	0.6-0.8 MPa							
3	Availability of cooling water	Available							
4	NaClO	> 20 L							
5	NaOH	> 20 L							
6	Plant Steam Pressure (on the header)	3-4 Bar							
7	Press. Diff. of 5 Micron Filter Circulation On	≤ 7 psi							
	Press. Diff. of 5 Micron Filter Idle/Standby	Monitoring Only							
8	UV Lamp Hour Meter	< 8500 h							
9	PW Temp. On the PW Tank	≤ 40 °C							
10	PW Temp. On the looping return 1st floor	Monitoring Only							
11	PW Temp. On the looping return 2nd floor	Monitoring Only							
12	PW Conductivity after EDI Circulation On	< 1.2 µ/cm							
	PW Conductivity after EDI Idle/Stanby	Monitoring Only							
13	PW Cond. On the looping return 1st floor	< 1.2 µ/cm							
14	PW Cond. On the looping return 2nd floor	< 1.2 µ/cm							
15	PW Loop 1 Flow Velocity	> 1 m/s							
16	PW Loop 2 Flow Velocity	> 1 m/s							
17	pH	7-9							
18	Soft Water total hardnees (once a day)	< 10 ppm							
19	Differential Pressure MMF	< 0.2 MPa							
20	Differential Pressure ACF	< 0.2 MPa							
21	Cleanliness of machine and room	Clean							

Notes : PW Conductivity dan PW Conductivity action limit : Sesuai Lampiran SOP/EBI/EN-015-L07 UV Lamp hour meter alert limit : 8000 h, UV Lamp hour meter action limit : 8500 h PW Loop Flow Velocity alert limit : 1.1 m/s, PW Loop Flow Velocity action limit : 1 m/s Pressure Diff. of 5 micron filter alert limit : 6 psi Pressure Diff. of 5 micron filter action limit : 7 psi Pembuatan NaOH dan NaClO mengacu pada SOP/EBI/EN-015. point 7.1.1.2 dan 7.1.1.3.	Operator : Shift 1 Shift 2 Shift 3 () () ()	Checked By : ()
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 PT ETANA BIOTECHNOLOGIES INDONESIA	LOGBOOK / LOGBOOK						Document No. : SOP/EBI/EN-015-F01H
	PEMANTAUAN HARIAN TERHADAP GENERATOR AIR MURNI DAN DISTRIBUSI SKIDNYA DAILY MONITORING OF PW GENERATOR AND THE DISTRIBUTION SKID						

Date :

No	Checklist Items	Standard Value	Time : (Shift 1, Shift 2 dan Shift 3)						Remark
			07:30	11:30	15:30	19:30	23:30	03:30	
1	Piping, Pneumatic hose, Valves	No Leaks							
2	Compressed Air Pressure	0.6-0.8 MPa							
3	Availability of cooling water	Available							
4	NaClO	> 20 L							
5	NaOH	> 20 L							
6	Plant Steam Pressure (on the header)	3-4 Bar							
7	Press. Diff. of 5 Micron Filter Circulation On	≤ 7 psi							
	Press. Diff. of 5 Micron Filter Idle/Standby	Monitoring Only							
8	UV Lamp Hour Meter	< 8500 h							
9	PW Temp. On the PW Tank	≤ 40 °C							
10	PW Temp. On the looping return 1st floor	Monitoring Only							
11	PW Temp. On the looping return 2nd floor	Monitoring Only							
12	PW Conductivity after EDI Circulation On	< 1.2 μ/cm							
	PW Conductivity after EDI Idle/Stanby	Monitoring Only							
13	PW Cond. On the looping return 1st floor	< 1.2 μ/cm							
14	PW Cond. On the looping return 2nd floor	< 1.2 μ/cm							
15	PW Loop 1 Flow Velocity	> 1 m/s							
16	PW Loop 2 Flow Velocity	> 1 m/s							
17	pH	7-9							
18	Soft Water total hardnees (once a day)	< 10 ppm							
19	Differential Pressure MMF	< 0.2 MPa							
20	Differential Pressure ACF	< 0.2 MPa							
21	Cleanliness of machine and room	Clean							

Notes : PW Conductivity dan PW Conductivity action limit : Sesuai Lampiran SOP/EBI/EN-015-L07 UV Lamp hour meter alert limit : 8000 h, UV Lamp hour meter action limit : 8500 h PW Loop Flow Velocity alert limit : 1.1 m/s, PW Loop Flow Velocity action limit : 1 m/s Pressure Diff. of 5 micron filter alert limit : 6 psi Pressure Diff. of 5 micron filter action limit : 7 psi Pembuatan NaOH dan NaClO mengacu pada SOP/EBI/EN-015. point 7.1.1.2 dan 7.1.1.3.	Operator : Shift 1 Shift 2 Shift 3 () () ()	Checked By : ()
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 PT ETANA BIOTECHNOLOGIES INDONESIA	LOGBOOK / LOGBOOK						Document No. : SOP/EBI/EN-015-F01H
	PEMANTAUAN HARIAN TERHADAP GENERATOR AIR MURNI DAN DISTRIBUSI SKIDNYA DAILY MONITORING OF PW GENERATOR AND THE DISTRIBUTION SKID						

Date :

No	Checklist Items	Standard Value	Time : (Shift 1, Shift 2 dan Shift 3)						Remark
			07:30	11:30	15:30	19:30	23:30	03:30	
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2	Compressed Air Pressure	0.6-0.8 MPa							
3	Availability of cooling water	Available							
4	NaClO	> 20 L							
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6	Plant Steam Pressure (on the header)	3-4 Bar							
7	Press. Diff. of 5 Micron Filter Circulation On	≤ 7 psi							
	Press. Diff. of 5 Micron Filter Idle/Standby	Monitoring Only							
8	UV Lamp Hour Meter	< 8500 h							
9	PW Temp. On the PW Tank	≤ 40 °C							
10	PW Temp. On the looping return 1st floor	Monitoring Only							
11	PW Temp. On the looping return 2nd floor	Monitoring Only							
12	PW Conductivity after EDI Circulation On	< 1.2 µ/cm							
	PW Conductivity after EDI Idle/Stanby	Monitoring Only							
13	PW Cond. On the looping return 1st floor	< 1.2 µ/cm							
14	PW Cond. On the looping return 2nd floor	< 1.2 µ/cm							
15	PW Loop 1 Flow Velocity	> 1 m/s							
16	PW Loop 2 Flow Velocity	> 1 m/s							
17	pH	7-9							
18	Soft Water total hardnees (once a day)	< 10 ppm							
19	Differential Pressure MMF	< 0.2 MPa							
20	Differential Pressure ACF	< 0.2 MPa							
21	Cleanliness of machine and room	Clean							
Notes : PW Conductivity dan PW Conductivity action limit : Sesuai Lampiran SOP/EBI/EN-015-L07 UV Lamp hour meter alert limit : 8000 h, UV Lamp hour meter action limit : 8500 h PW Loop Flow Velocity alert limit : 1.1 m/s, PW Loop Flow Velocity action limit : 1 m/s Pressure Diff. of 5 micron filter alert limit : 6 psi Pressure Diff. of 5 micron filter action limit : 7 psi Pembuatan NaOH dan NaClO mengacu pada SOP/EBI/EN-015. point 7.1.1.2 dan 7.1.1.3.			Operator : Shift 1 Shift 2 Shift 3 () () ()						Checked By : ()

 PT ETANA BIOTECHNOLOGIES INDONESIA	LOGBOOK / LOGBOOK						Document No. : SOP/EBI/EN-015-F01H
	PEMANTAUAN HARIAN TERHADAP GENERATOR AIR MURNI DAN DISTRIBUSI SKIDNYA DAILY MONITORING OF PW GENERATOR AND THE DISTRIBUTION SKID						

Date :

No	Checklist Items	Standard Value	Time : (Shift 1, Shift 2 dan Shift 3)						Remark
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3	Availability of cooling water	Available							
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6	Plant Steam Pressure (on the header)	3-4 Bar							
7	Press. Diff. of 5 Micron Filter Circulation On	≤ 7 psi							
	Press. Diff. of 5 Micron Filter Idle/Standby	Monitoring Only							
8	UV Lamp Hour Meter	< 8500 h							
9	PW Temp. On the PW Tank	≤ 40 °C							
10	PW Temp. On the looping return 1st floor	Monitoring Only							
11	PW Temp. On the looping return 2nd floor	Monitoring Only							
12	PW Conductivity after EDI Circulation On	< 1.2 µ/cm							
	PW Conductivity after EDI Idle/Stanby	Monitoring Only							
13	PW Cond. On the looping return 1st floor	< 1.2 µ/cm							
14	PW Cond. On the looping return 2nd floor	< 1.2 µ/cm							
15	PW Loop 1 Flow Velocity	> 1 m/s							
16	PW Loop 2 Flow Velocity	> 1 m/s							
17	pH	7-9							
18	Soft Water total hardnees (once a day)	< 10 ppm							
19	Differential Pressure MMF	< 0.2 MPa							
20	Differential Pressure ACF	< 0.2 MPa							
21	Cleanliness of machine and room	Clean							

Notes : PW Conductivity dan PW Conductivity action limit : Sesuai Lampiran SOP/EBI/EN-015-L07 UV Lamp hour meter alert limit : 8000 h, UV Lamp hour meter action limit : 8500 h PW Loop Flow Velocity alert limit : 1.1 m/s, PW Loop Flow Velocity action limit : 1 m/s Pressure Diff. of 5 micron filter alert limit : 6 psi Pressure Diff. of 5 micron filter action limit : 7 psi Pembuatan NaOH dan NaClO mengacu pada SOP/EBI/EN-015. point 7.1.1.2 dan 7.1.1.3.	Operator : Shift 1 Shift 2 Shift 3 () () ()	Checked By : ()
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 PT ETANA BIOTECHNOLOGIES INDONESIA	LOGBOOK / LOGBOOK						Document No. : SOP/EBI/EN-015-F01H
	PEMANTAUAN HARIAN TERHADAP GENERATOR AIR MURNI DAN DISTRIBUSI SKIDNYA DAILY MONITORING OF PW GENERATOR AND THE DISTRIBUTION SKID						

Date :

No	Checklist Items	Standard Value	Time : (Shift 1, Shift 2 dan Shift 3)						Remark
			07:30	11:30	15:30	19:30	23:30	03:30	
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	Press. Diff. of 5 Micron Filter Idle/Standby	Monitoring Only							
8	UV Lamp Hour Meter	< 8500 h							
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12	PW Conductivity after EDI Circulation On	< 1.2 µ/cm							
	PW Conductivity after EDI Idle/Stanby	Monitoring Only							
13	PW Cond. On the looping return 1st floor	< 1.2 µ/cm							
14	PW Cond. On the looping return 2nd floor	< 1.2 µ/cm							
15	PW Loop 1 Flow Velocity	> 1 m/s							
16	PW Loop 2 Flow Velocity	> 1 m/s							
17	pH	7-9							
18	Soft Water total hardnees (once a day)	< 10 ppm							
19	Differential Pressure MMF	< 0.2 MPa							
20	Differential Pressure ACF	< 0.2 MPa							
21	Cleanliness of machine and room	Clean							

Notes :

PW Conductivity dan PW Conductivity action limit : Sesuai Lampiran SOP/EBI/EN-015-L07
 UV Lamp hour meter alert limit : 8000 h, UV Lamp hour meter action limit : 8500 h
 PW Loop Flow Velocity alert limit : 1.1 m/s, PW Loop Flow Velocity action limit : 1 m/s
 Pressure Diff. of 5 micron filter alert limit : 6 psi
 Pressure Diff. of 5 micron filter action limit : 7 psi
 Pembuatan NaOH dan NaClO mengacu pada SOP/EBI/EN-015. point 7.1.1.2 dan 7.1.1.3.

Operator :

Shift 1

Shift 2

Shift 3

Checked By :

() () ()

()

 PT ETANA BIOTECHNOLOGIES INDONESIA	LOGBOOK / LOGBOOK						Document No. : SOP/EBI/EN-015-F01H
	PEMANTAUAN HARIAN TERHADAP GENERATOR AIR MURNI DAN DISTRIBUSI SKIDNYA DAILY MONITORING OF PW GENERATOR AND THE DISTRIBUTION SKID						

Date :

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	Press. Diff. of 5 Micron Filter Idle/Standby	Monitoring Only							
8	UV Lamp Hour Meter	< 8500 h							
9	PW Temp. On the PW Tank	≤ 40 °C							
10	PW Temp. On the looping return 1st floor	Monitoring Only							
11	PW Temp. On the looping return 2nd floor	Monitoring Only							
12	PW Conductivity after EDI Circulation On	< 1.2 µ/cm							
	PW Conductivity after EDI Idle/Stanby	Monitoring Only							
13	PW Cond. On the looping return 1st floor	< 1.2 µ/cm							
14	PW Cond. On the looping return 2nd floor	< 1.2 µ/cm							
15	PW Loop 1 Flow Velocity	> 1 m/s							
16	PW Loop 2 Flow Velocity	> 1 m/s							
17	pH	7-9							
18	Soft Water total hardnees (once a day)	< 10 ppm							
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Notes : PW Conductivity dan PW Conductivity action limit : Sesuai Lampiran SOP/EBI/EN-015-L07 UV Lamp hour meter alert limit : 8000 h, UV Lamp hour meter action limit : 8500 h PW Loop Flow Velocity alert limit : 1.1 m/s, PW Loop Flow Velocity action limit : 1 m/s Pressure Diff. of 5 micron filter alert limit : 6 psi Pressure Diff. of 5 micron filter action limit : 7 psi Pembuatan NaOH dan NaClO mengacu pada SOP/EBI/EN-015. point 7.1.1.2 dan 7.1.1.3.	Operator : Shift 1 Shift 2 Shift 3 () () ()	Checked By : ()
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 PT ETANA BIOTECHNOLOGIES INDONESIA	LOGBOOK / LOGBOOK						Document No. : SOP/EBI/EN-015-F01H
	PEMANTAUAN HARIAN TERHADAP GENERATOR AIR MURNI DAN DISTRIBUSI SKIDNYA DAILY MONITORING OF PW GENERATOR AND THE DISTRIBUTION SKID						

Date :

No	Checklist Items	Standard Value	Time : (Shift 1, Shift 2 dan Shift 3)						Remark
			07:30	11:30	15:30	19:30	23:30	03:30	
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3	Availability of cooling water	Available							
4	NaClO	> 20 L							
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6	Plant Steam Pressure (on the header)	3-4 Bar							
7	Press. Diff. of 5 Micron Filter Circulation On	≤ 7 psi							
	Press. Diff. of 5 Micron Filter Idle/Standby	Monitoring Only							
8	UV Lamp Hour Meter	< 8500 h							
9	PW Temp. On the PW Tank	≤ 40 °C							
10	PW Temp. On the looping return 1st floor	Monitoring Only							
11	PW Temp. On the looping return 2nd floor	Monitoring Only							
12	PW Conductivity after EDI Circulation On	< 1.2 µ/cm							
	PW Conductivity after EDI Idle/Stanby	Monitoring Only							
13	PW Cond. On the looping return 1st floor	< 1.2 µ/cm							
14	PW Cond. On the looping return 2nd floor	< 1.2 µ/cm							
15	PW Loop 1 Flow Velocity	> 1 m/s							
16	PW Loop 2 Flow Velocity	> 1 m/s							
17	pH	7-9							
18	Soft Water total hardnees (once a day)	< 10 ppm							
19	Differential Pressure MMF	< 0.2 MPa							
20	Differential Pressure ACF	< 0.2 MPa							
21	Cleanliness of machine and room	Clean							

Notes :

PW Conductivity dan PW Conductivity action limit : Sesuai Lampiran SOP/EBI/EN-015-L07
 UV Lamp hour meter alert limit : 8000 h, UV Lamp hour meter action limit : 8500 h
 PW Loop Flow Velocity alert limit : 1.1 m/s, PW Loop Flow Velocity action limit : 1 m/s
 Pressure Diff. of 5 micron filter alert limit : 6 psi
 Pressure Diff. of 5 micron filter action limit : 7 psi
 Pembuatan NaOH dan NaClO mengacu pada SOP/EBI/EN-015. point 7.1.1.2 dan 7.1.1.3.

Operator :

Shift 1

Shift 2

Shift 3

Checked By :

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 PT ETANA BIOTECHNOLOGIES INDONESIA	LOGBOOK / LOGBOOK						Document No. : SOP/EBI/EN-015-F01H
	PEMANTAUAN HARIAN TERHADAP GENERATOR AIR MURNI DAN DISTRIBUSI SKIDNYA DAILY MONITORING OF PW GENERATOR AND THE DISTRIBUTION SKID						

Date :

No	Checklist Items	Standard Value	Time : (Shift 1, Shift 2 dan Shift 3)						Remark
			07:30	11:30	15:30	19:30	23:30	03:30	
1	Piping, Pneumatic hose, Valves	No Leaks							
2	Compressed Air Pressure	0.6-0.8 MPa							
3	Availability of cooling water	Available							
4	NaClO	> 20 L							
5	NaOH	> 20 L							
6	Plant Steam Pressure (on the header)	3-4 Bar							
7	Press. Diff. of 5 Micron Filter Circulation On	≤ 7 psi							
	Press. Diff. of 5 Micron Filter Idle/Standby	Monitoring Only							
8	UV Lamp Hour Meter	< 8500 h							
9	PW Temp. On the PW Tank	≤ 40 °C							
10	PW Temp. On the looping return 1st floor	Monitoring Only							
11	PW Temp. On the looping return 2nd floor	Monitoring Only							
12	PW Conductivity after EDI Circulation On	< 1.2 µ/cm							
	PW Conductivity after EDI Idle/Stanby	Monitoring Only							
13	PW Cond. On the looping return 1st floor	< 1.2 µ/cm							
14	PW Cond. On the looping return 2nd floor	< 1.2 µ/cm							
15	PW Loop 1 Flow Velocity	> 1 m/s							
16	PW Loop 2 Flow Velocity	> 1 m/s							
17	pH	7-9							
18	Soft Water total hardnees (once a day)	< 10 ppm							
19	Differential Pressure MMF	< 0.2 MPa							
20	Differential Pressure ACF	< 0.2 MPa							
21	Cleanliness of machine and room	Clean							

Notes :

PW Conductivity dan PW Conductivity action limit : Sesuai Lampiran SOP/EBI/EN-015-L07
 UV Lamp hour meter alert limit : 8000 h, UV Lamp hour meter action limit : 8500 h
 PW Loop Flow Velocity alert limit : 1.1 m/s, PW Loop Flow Velocity action limit : 1 m/s
 Pressure Diff. of 5 micron filter alert limit : 6 psi
 Pressure Diff. of 5 micron filter action limit : 7 psi
 Pembuatan NaOH dan NaClO mengacu pada SOP/EBI/EN-015. point 7.1.1.2 dan 7.1.1.3.

Operator :

Shift 1

Shift 2

Shift 3

Checked By :

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 PT ETANA BIOTECHNOLOGIES INDONESIA	LOGBOOK / LOGBOOK						Document No. : SOP/EBI/EN-015-F01H
	PEMANTAUAN HARIAN TERHADAP GENERATOR AIR MURNI DAN DISTRIBUSI SKIDNYA DAILY MONITORING OF PW GENERATOR AND THE DISTRIBUTION SKID						

Date :

No	Checklist Items	Standard Value	Time : (Shift 1, Shift 2 dan Shift 3)						Remark
			07:30	11:30	15:30	19:30	23:30	03:30	
1	Piping, Pneumatic hose, Valves	No Leaks							
2	Compressed Air Pressure	0.6-0.8 MPa							
3	Availability of cooling water	Available							
4	NaClO	> 20 L							
5	NaOH	> 20 L							
6	Plant Steam Pressure (on the header)	3-4 Bar							
7	Press. Diff. of 5 Micron Filter Circulation On	≤ 7 psi							
	Press. Diff. of 5 Micron Filter Idle/Standby	Monitoring Only							
8	UV Lamp Hour Meter	< 8500 h							
9	PW Temp. On the PW Tank	≤ 40 °C							
10	PW Temp. On the looping return 1st floor	Monitoring Only							
11	PW Temp. On the looping return 2nd floor	Monitoring Only							
12	PW Conductivity after EDI Circulation On	< 1.2 µ/cm							
	PW Conductivity after EDI Idle/Stanby	Monitoring Only							
13	PW Cond. On the looping return 1st floor	< 1.2 µ/cm							
14	PW Cond. On the looping return 2nd floor	< 1.2 µ/cm							
15	PW Loop 1 Flow Velocity	> 1 m/s							
16	PW Loop 2 Flow Velocity	> 1 m/s							
17	pH	7-9							
18	Soft Water total hardnees (once a day)	< 10 ppm							
19	Differential Pressure MMF	< 0.2 MPa							
20	Differential Pressure ACF	< 0.2 MPa							
21	Cleanliness of machine and room	Clean							
Notes : PW Conductivity dan PW Conductivity action limit : Sesuai Lampiran SOP/EBI/EN-015-L07 UV Lamp hour meter alert limit : 8000 h, UV Lamp hour meter action limit : 8500 h PW Loop Flow Velocity alert limit : 1.1 m/s, PW Loop Flow Velocity action limit : 1 m/s Pressure Diff. of 5 micron filter alert limit : 6 psi Pressure Diff. of 5 micron filter action limit : 7 psi Pembuatan NaOH dan NaClO mengacu pada SOP/EBI/EN-015. point 7.1.1.2 dan 7.1.1.3.			Operator : Shift 1 Shift 2 Shift 3 () () ()						Checked By : ()

 PT ETANA BIOTECHNOLOGIES INDONESIA	LOGBOOK / LOGBOOK						Document No. : SOP/EBI/EN-015-F01H
	PEMANTAUAN HARIAN TERHADAP GENERATOR AIR MURNI DAN DISTRIBUSI SKIDNYA DAILY MONITORING OF PW GENERATOR AND THE DISTRIBUTION SKID						

Date :

No	Checklist Items	Standard Value	Time : (Shift 1, Shift 2 dan Shift 3)						Remark
			07:30	11:30	15:30	19:30	23:30	03:30	
1	Piping, Pneumatic hose, Valves	No Leaks							
2	Compressed Air Pressure	0.6-0.8 MPa							
3	Availability of cooling water	Available							
4	NaClO	> 20 L							
5	NaOH	> 20 L							
6	Plant Steam Pressure (on the header)	3-4 Bar							
7	Press. Diff. of 5 Micron Filter Circulation On	≤ 7 psi							
	Press. Diff. of 5 Micron Filter Idle/Standby	Monitoring Only							
8	UV Lamp Hour Meter	< 8500 h							
9	PW Temp. On the PW Tank	≤ 40 °C							
10	PW Temp. On the looping return 1st floor	Monitoring Only							
11	PW Temp. On the looping return 2nd floor	Monitoring Only							
12	PW Conductivity after EDI Circulation On	< 1.2 µ/cm							
	PW Conductivity after EDI Idle/Stanby	Monitoring Only							
13	PW Cond. On the looping return 1st floor	< 1.2 µ/cm							
14	PW Cond. On the looping return 2nd floor	< 1.2 µ/cm							
15	PW Loop 1 Flow Velocity	> 1 m/s							
16	PW Loop 2 Flow Velocity	> 1 m/s							
17	pH	7-9							
18	Soft Water total hardnees (once a day)	< 10 ppm							
19	Differential Pressure MMF	< 0.2 MPa							
20	Differential Pressure ACF	< 0.2 MPa							
21	Cleanliness of machine and room	Clean							

Notes : PW Conductivity dan PW Conductivity action limit : Sesuai Lampiran SOP/EBI/EN-015-L07 UV Lamp hour meter alert limit : 8000 h, UV Lamp hour meter action limit : 8500 h PW Loop Flow Velocity alert limit : 1.1 m/s, PW Loop Flow Velocity action limit : 1 m/s Pressure Diff. of 5 micron filter alert limit : 6 psi Pressure Diff. of 5 micron filter action limit : 7 psi Pembuatan NaOH dan NaClO mengacu pada SOP/EBI/EN-015. point 7.1.1.2 dan 7.1.1.3.	Operator : Shift 1 Shift 2 Shift 3 () () ()	Checked By : ()
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 PT ETANA BIOTECHNOLOGIES INDONESIA	LOGBOOK / LOGBOOK						Document No. : SOP/EBI/EN-015-F01H
	PEMANTAUAN HARIAN TERHADAP GENERATOR AIR MURNI DAN DISTRIBUSI SKIDNYA DAILY MONITORING OF PW GENERATOR AND THE DISTRIBUTION SKID						

Date :

No	Checklist Items	Standard Value	Time : (Shift 1, Shift 2 dan Shift 3)						Remark
			07:30	11:30	15:30	19:30	23:30	03:30	
1	Piping, Pneumatic hose, Valves	No Leaks							
2	Compressed Air Pressure	0.6-0.8 MPa							
3	Availability of cooling water	Available							
4	NaClO	> 20 L							
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6	Plant Steam Pressure (on the header)	3-4 Bar							
7	Press. Diff. of 5 Micron Filter Circulation On	≤ 7 psi							
	Press. Diff. of 5 Micron Filter Idle/Standby	Monitoring Only							
8	UV Lamp Hour Meter	< 8500 h							
9	PW Temp. On the PW Tank	≤ 40 °C							
10	PW Temp. On the looping return 1st floor	Monitoring Only							
11	PW Temp. On the looping return 2nd floor	Monitoring Only							
12	PW Conductivity after EDI Circulation On	< 1.2 µ/cm							
	PW Conductivity after EDI Idle/Stanby	Monitoring Only							
13	PW Cond. On the looping return 1st floor	< 1.2 µ/cm							
14	PW Cond. On the looping return 2nd floor	< 1.2 µ/cm							
15	PW Loop 1 Flow Velocity	> 1 m/s							
16	PW Loop 2 Flow Velocity	> 1 m/s							
17	pH	7-9							
18	Soft Water total hardnees (once a day)	< 10 ppm							
19	Differential Pressure MMF	< 0.2 MPa							
20	Differential Pressure ACF	< 0.2 MPa							
21	Cleanliness of machine and room	Clean							

Notes : PW Conductivity dan PW Conductivity action limit : Sesuai Lampiran SOP/EBI/EN-015-L07 UV Lamp hour meter alert limit : 8000 h, UV Lamp hour meter action limit : 8500 h PW Loop Flow Velocity alert limit : 1.1 m/s, PW Loop Flow Velocity action limit : 1 m/s Pressure Diff. of 5 micron filter alert limit : 6 psi Pressure Diff. of 5 micron filter action limit : 7 psi Pembuatan NaOH dan NaClO mengacu pada SOP/EBI/EN-015. point 7.1.1.2 dan 7.1.1.3.	Operator : Shift 1 Shift 2 Shift 3 () () ()	Checked By : ()
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 PT ETANA BIOTECHNOLOGIES INDONESIA	LOGBOOK / LOGBOOK		Document No. : SOP/EBI/EN-015-F01H
	PEMANTAUAN HARIAN TERHADAP GENERATOR AIR MURNI DAN DISTRIBUSI SKIDNYA DAILY MONITORING OF PW GENERATOR AND THE DISTRIBUTION SKID		

Date :

No	Checklist Items	Standard Value	Time : (Shift 1, Shift 2 dan Shift 3)						Remark
			07:30	11:30	15:30	19:30	23:30	03:30	
1	Piping, Pneumatic hose, Valves	No Leaks							
2	Compressed Air Pressure	0.6-0.8 MPa							
3	Availability of cooling water	Available							
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7	Press. Diff. of 5 Micron Filter Circulation On	≤ 7 psi							
	Press. Diff. of 5 Micron Filter Idle/Standby	Monitoring Only							
8	UV Lamp Hour Meter	< 8500 h							
9	PW Temp. On the PW Tank	≤ 40 °C							
10	PW Temp. On the looping return 1st floor	Monitoring Only							
11	PW Temp. On the looping return 2nd floor	Monitoring Only							
12	PW Conductivity after EDI Circulation On	< 1.2 µ/cm							
	PW Conductivity after EDI Idle/Stanby	Monitoring Only							
13	PW Cond. On the looping return 1st floor	< 1.2 µ/cm							
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17	pH	7-9							
18	Soft Water total hardnees (once a day)	< 10 ppm							
19	Differential Pressure MMF	< 0.2 MPa							
20	Differential Pressure ACF	< 0.2 MPa							
21	Cleanliness of machine and room	Clean							

Notes : PW Conductivity dan PW Conductivity action limit : Sesuai Lampiran SOP/EBI/EN-015-L07 UV Lamp hour meter alert limit : 8000 h, UV Lamp hour meter action limit : 8500 h PW Loop Flow Velocity alert limit : 1.1 m/s, PW Loop Flow Velocity action limit : 1 m/s Pressure Diff. of 5 micron filter alert limit : 6 psi Pressure Diff. of 5 micron filter action limit : 7 psi Pembuatan NaOH dan NaClO mengacu pada SOP/EBI/EN-015. point 7.1.1.2 dan 7.1.1.3.	Operator : Shift 1 Shift 2 Shift 3 () () ()	Checked By : ()
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 PT ETANA BIOTECHNOLOGIES INDONESIA	LOGBOOK / LOGBOOK						Document No. : SOP/EBI/EN-015-F01H
	PEMANTAUAN HARIAN TERHADAP GENERATOR AIR MURNI DAN DISTRIBUSI SKIDNYA DAILY MONITORING OF PW GENERATOR AND THE DISTRIBUTION SKID						

Date :

No	Checklist Items	Standard Value	Time : (Shift 1, Shift 2 dan Shift 3)						Remark
			07:30	11:30	15:30	19:30	23:30	03:30	
1	Piping, Pneumatic hose, Valves	No Leaks							
2	Compressed Air Pressure	0.6-0.8 MPa							
3	Availability of cooling water	Available							
4	NaClO	> 20 L							
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6	Plant Steam Pressure (on the header)	3-4 Bar							
7	Press. Diff. of 5 Micron Filter Circulation On	≤ 7 psi							
	Press. Diff. of 5 Micron Filter Idle/Standby	Monitoring Only							
8	UV Lamp Hour Meter	< 8500 h							
9	PW Temp. On the PW Tank	≤ 40 °C							
10	PW Temp. On the looping return 1st floor	Monitoring Only							
11	PW Temp. On the looping return 2nd floor	Monitoring Only							
12	PW Conductivity after EDI Circulation On	< 1.2 µ/cm							
	PW Conductivity after EDI Idle/Stanby	Monitoring Only							
13	PW Cond. On the looping return 1st floor	< 1.2 µ/cm							
14	PW Cond. On the looping return 2nd floor	< 1.2 µ/cm							
15	PW Loop 1 Flow Velocity	> 1 m/s							
16	PW Loop 2 Flow Velocity	> 1 m/s							
17	pH	7-9							
18	Soft Water total hardnees (once a day)	< 10 ppm							
19	Differential Pressure MMF	< 0.2 MPa							
20	Differential Pressure ACF	< 0.2 MPa							
21	Cleanliness of machine and room	Clean							

Notes : PW Conductivity dan PW Conductivity action limit : Sesuai Lampiran SOP/EBI/EN-015-L07 UV Lamp hour meter alert limit : 8000 h, UV Lamp hour meter action limit : 8500 h PW Loop Flow Velocity alert limit : 1.1 m/s, PW Loop Flow Velocity action limit : 1 m/s Pressure Diff. of 5 micron filter alert limit : 6 psi Pressure Diff. of 5 micron filter action limit : 7 psi Pembuatan NaOH dan NaClO mengacu pada SOP/EBI/EN-015. point 7.1.1.2 dan 7.1.1.3.	Operator : Shift 1 Shift 2 Shift 3 () () ()	Checked By : ()
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 PT ETANA BIOTECHNOLOGIES INDONESIA	LOGBOOK / LOGBOOK						Document No. : SOP/EBI/EN-015-F01H
	PEMANTAUAN HARIAN TERHADAP GENERATOR AIR MURNI DAN DISTRIBUSI SKIDNYA DAILY MONITORING OF PW GENERATOR AND THE DISTRIBUTION SKID						

Date :

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			07:30	11:30	15:30	19:30	23:30	03:30	
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2	Compressed Air Pressure	0.6-0.8 MPa							
3	Availability of cooling water	Available							
4	NaClO	> 20 L							
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6	Plant Steam Pressure (on the header)	3-4 Bar							
7	Press. Diff. of 5 Micron Filter Circulation On	≤ 7 psi							
	Press. Diff. of 5 Micron Filter Idle/Standby	Monitoring Only							
8	UV Lamp Hour Meter	< 8500 h							
9	PW Temp. On the PW Tank	≤ 40 °C							
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13	PW Cond. On the looping return 1st floor	< 1.2 µ/cm							
14	PW Cond. On the looping return 2nd floor	< 1.2 µ/cm							
15	PW Loop 1 Flow Velocity	> 1 m/s							
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17	pH	7-9							
18	Soft Water total hardnees (once a day)	< 10 ppm							
19	Differential Pressure MMF	< 0.2 MPa							
20	Differential Pressure ACF	< 0.2 MPa							
21	Cleanliness of machine and room	Clean							

Notes :

PW Conductivity dan PW Conductivity action limit : Sesuai Lampiran SOP/EBI/EN-015-L07

UV Lamp hour meter alert limit : 8000 h, UV Lamp hour meter action limit : 8500 h

PW Loop Flow Velocity alert limit : 1.1 m/s, PW Loop Flow Velocity action limit : 1 m/s

Pressure Diff. of 5 micron filter alert limit : 6 psi

Pressure Diff. of 5 micron filter action limit : 7 psi

Pembuatan NaOH dan NaClO mengacu pada SOP/EBI/EN-015. point 7.1.1.2 dan 7.1.1.3.

Operator :

Shift 1	Shift 2	Shift 3
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Checked By :

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 PT ETANA BIOTECHNOLOGIES INDONESIA	LOGBOOK / LOGBOOK						Document No. : SOP/EBI/EN-015-F01H
	PEMANTAUAN HARIAN TERHADAP GENERATOR AIR MURNI DAN DISTRIBUSI SKIDNYA DAILY MONITORING OF PW GENERATOR AND THE DISTRIBUTION SKID						

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12	PW Conductivity after EDI Circulation On	< 1.2 µ/cm							
	PW Conductivity after EDI Idle/Stanby	Monitoring Only							
13	PW Cond. On the looping return 1st floor	< 1.2 µ/cm							
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15	PW Loop 1 Flow Velocity	> 1 m/s							
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17	pH	7-9							
18	Soft Water total hardnees (once a day)	< 10 ppm							
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Notes : PW Conductivity dan PW Conductivity action limit : Sesuai Lampiran SOP/EBI/EN-015-L07 UV Lamp hour meter alert limit : 8000 h, UV Lamp hour meter action limit : 8500 h PW Loop Flow Velocity alert limit : 1.1 m/s, PW Loop Flow Velocity action limit : 1 m/s Pressure Diff. of 5 micron filter alert limit : 6 psi Pressure Diff. of 5 micron filter action limit : 7 psi Pembuatan NaOH dan NaClO mengacu pada SOP/EBI/EN-015. point 7.1.1.2 dan 7.1.1.3.	Operator : Shift 1 Shift 2 Shift 3 () () ()	Checked By : ()
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 PT ETANA BIOTECHNOLOGIES INDONESIA	LOGBOOK / LOGBOOK						Document No. : SOP/EBI/EN-015-F01H
	PEMANTAUAN HARIAN TERHADAP GENERATOR AIR MURNI DAN DISTRIBUSI SKIDNYA DAILY MONITORING OF PW GENERATOR AND THE DISTRIBUTION SKID						

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15	PW Loop 1 Flow Velocity	> 1 m/s							
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	PEMANTAUAN HARIAN TERHADAP GENERATOR AIR MURNI DAN DISTRIBUSI SKIDNYA DAILY MONITORING OF PW GENERATOR AND THE DISTRIBUTION SKID						

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15	PW Loop 1 Flow Velocity	> 1 m/s							
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 PT ETANA BIOTECHNOLOGIES INDONESIA	LOGBOOK / LOGBOOK						Document No. : SOP/EBI/EN-015-F01H
	PEMANTAUAN HARIAN TERHADAP GENERATOR AIR MURNI DAN DISTRIBUSI SKIDNYA DAILY MONITORING OF PW GENERATOR AND THE DISTRIBUTION SKID						

Date :

No	Checklist Items	Standard Value	Time : (Shift 1, Shift 2 dan Shift 3)						Remark
			07:30	11:30	15:30	19:30	23:30	03:30	
1	Piping, Pneumatic hose, Valves	No Leaks							
2	Compressed Air Pressure	0.6-0.8 MPa							
3	Availability of cooling water	Available							
4	NaClO	> 20 L							
5	NaOH	> 20 L							
6	Plant Steam Pressure (on the header)	3-4 Bar							
7	Press. Diff. of 5 Micron Filter Circulation On	≤ 7 psi							
	Press. Diff. of 5 Micron Filter Idle/Standby	Monitoring Only							
8	UV Lamp Hour Meter	< 8500 h							
9	PW Temp. On the PW Tank	≤ 40 °C							
10	PW Temp. On the looping return 1st floor	Monitoring Only							
11	PW Temp. On the looping return 2nd floor	Monitoring Only							
12	PW Conductivity after EDI Circulation On	< 1.2 µ/cm							
	PW Conductivity after EDI Idle/Stanby	Monitoring Only							
13	PW Cond. On the looping return 1st floor	< 1.2 µ/cm							
14	PW Cond. On the looping return 2nd floor	< 1.2 µ/cm							
15	PW Loop 1 Flow Velocity	> 1 m/s							
16	PW Loop 2 Flow Velocity	> 1 m/s							
17	pH	7-9							
18	Soft Water total hardnees (once a day)	< 10 ppm							
19	Differential Pressure MMF	< 0.2 MPa							
20	Differential Pressure ACF	< 0.2 MPa							
21	Cleanliness of machine and room	Clean							

Notes : PW Conductivity dan PW Conductivity action limit : Sesuai Lampiran SOP/EBI/EN-015-L07 UV Lamp hour meter alert limit : 8000 h, UV Lamp hour meter action limit : 8500 h PW Loop Flow Velocity alert limit : 1.1 m/s, PW Loop Flow Velocity action limit : 1 m/s Pressure Diff. of 5 micron filter alert limit : 6 psi Pressure Diff. of 5 micron filter action limit : 7 psi Pembuatan NaOH dan NaClO mengacu pada SOP/EBI/EN-015. point 7.1.1.2 dan 7.1.1.3.	Operator : Shift 1 Shift 2 Shift 3 () () ()	Checked By : ()
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 PT ETANA BIOTECHNOLOGIES INDONESIA	LOGBOOK / LOGBOOK						Document No. : SOP/EBI/EN-015-F01H
	PEMANTAUAN HARIAN TERHADAP GENERATOR AIR MURNI DAN DISTRIBUSI SKIDNYA DAILY MONITORING OF PW GENERATOR AND THE DISTRIBUTION SKID						

Date :

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7	Press. Diff. of 5 Micron Filter Circulation On	≤ 7 psi							
	Press. Diff. of 5 Micron Filter Idle/Standby	Monitoring Only							
8	UV Lamp Hour Meter	< 8500 h							
9	PW Temp. On the PW Tank	≤ 40 °C							
10	PW Temp. On the looping return 1st floor	Monitoring Only							
11	PW Temp. On the looping return 2nd floor	Monitoring Only							
12	PW Conductivity after EDI Circulation On	< 1.2 μ/cm							
	PW Conductivity after EDI Idle/Stanby	Monitoring Only							
13	PW Cond. On the looping return 1st floor	< 1.2 μ/cm							
14	PW Cond. On the looping return 2nd floor	< 1.2 μ/cm							
15	PW Loop 1 Flow Velocity	> 1 m/s							
16	PW Loop 2 Flow Velocity	> 1 m/s							
17	pH	7-9							
18	Soft Water total hardnees (once a day)	< 10 ppm							
19	Differential Pressure MMF	< 0.2 MPa							
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21	Cleanliness of machine and room	Clean							

Notes : PW Conductivity dan PW Conductivity action limit : Sesuai Lampiran SOP/EBI/EN-015-L07 UV Lamp hour meter alert limit : 8000 h, UV Lamp hour meter action limit : 8500 h PW Loop Flow Velocity alert limit : 1.1 m/s, PW Loop Flow Velocity action limit : 1 m/s Pressure Diff. of 5 micron filter alert limit : 6 psi Pressure Diff. of 5 micron filter action limit : 7 psi Pembuatan NaOH dan NaClO mengacu pada SOP/EBI/EN-015. point 7.1.1.2 dan 7.1.1.3.	Operator : Shift 1 Shift 2 Shift 3 () () ()	Checked By : ()
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20	Differential Pressure ACF	< 0.2 MPa							
21	Cleanliness of machine and room	Clean							

Notes : PW Conductivity dan PW Conductivity action limit : Sesuai Lampiran SOP/EBI/EN-015-L07 UV Lamp hour meter alert limit : 8000 h, UV Lamp hour meter action limit : 8500 h PW Loop Flow Velocity alert limit : 1.1 m/s, PW Loop Flow Velocity action limit : 1 m/s Pressure Diff. of 5 micron filter alert limit : 6 psi Pressure Diff. of 5 micron filter action limit : 7 psi Pembuatan NaOH dan NaClO mengacu pada SOP/EBI/EN-015. point 7.1.1.2 dan 7.1.1.3.	Operator : Shift 1 Shift 2 Shift 3 () () ()	Checked By : ()
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 PT ETANA BIOTECHNOLOGIES INDONESIA	LOGBOOK / LOGBOOK						Document No. : SOP/EBI/EN-015-F01H
	PEMANTAUAN HARIAN TERHADAP GENERATOR AIR MURNI DAN DISTRIBUSI SKIDNYA DAILY MONITORING OF PW GENERATOR AND THE DISTRIBUTION SKID						

Date :

No	Checklist Items	Standard Value	Time : (Shift 1, Shift 2 dan Shift 3)						Remark
			07:30	11:30	15:30	19:30	23:30	03:30	
1	Piping, Pneumatic hose, Valves	No Leaks							
2	Compressed Air Pressure	0.6-0.8 MPa							
3	Availability of cooling water	Available							
4	NaClO	> 20 L							
5	NaOH	> 20 L							
6	Plant Steam Pressure (on the header)	3-4 Bar							
7	Press. Diff. of 5 Micron Filter Circulation On	≤ 7 psi							
	Press. Diff. of 5 Micron Filter Idle/Standby	Monitoring Only							
8	UV Lamp Hour Meter	< 8500 h							
9	PW Temp. On the PW Tank	≤ 40 °C							
10	PW Temp. On the looping return 1st floor	Monitoring Only							
11	PW Temp. On the looping return 2nd floor	Monitoring Only							
12	PW Conductivity after EDI Circulation On	< 1.2 µ/cm							
	PW Conductivity after EDI Idle/Stanby	Monitoring Only							
13	PW Cond. On the looping return 1st floor	< 1.2 µ/cm							
14	PW Cond. On the looping return 2nd floor	< 1.2 µ/cm							
15	PW Loop 1 Flow Velocity	> 1 m/s							
16	PW Loop 2 Flow Velocity	> 1 m/s							
17	pH	7-9							
18	Soft Water total hardnees (once a day)	< 10 ppm							
19	Differential Pressure MMF	< 0.2 MPa							
20	Differential Pressure ACF	< 0.2 MPa							
21	Cleanliness of machine and room	Clean							

Notes : PW Conductivity dan PW Conductivity action limit : Sesuai Lampiran SOP/EBI/EN-015-L07 UV Lamp hour meter alert limit : 8000 h, UV Lamp hour meter action limit : 8500 h PW Loop Flow Velocity alert limit : 1.1 m/s, PW Loop Flow Velocity action limit : 1 m/s Pressure Diff. of 5 micron filter alert limit : 6 psi Pressure Diff. of 5 micron filter action limit : 7 psi Pembuatan NaOH dan NaClO mengacu pada SOP/EBI/EN-015. point 7.1.1.2 dan 7.1.1.3.	Operator : Shift 1 Shift 2 Shift 3 () () ()	Checked By : ()
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 PT ETANA BIOTECHNOLOGIES INDONESIA	LOGBOOK / LOGBOOK		Document No. : SOP/EBI/EN-015-F01H
	PEMANTAUAN HARIAN TERHADAP GENERATOR AIR MURNI DAN DISTRIBUSI SKIDNYA DAILY MONITORING OF PW GENERATOR AND THE DISTRIBUTION SKID		

Date :

No	Checklist Items	Standard Value	Time : (Shift 1, Shift 2 dan Shift 3)						Remark
			07:30	11:30	15:30	19:30	23:30	03:30	
1	Piping, Pneumatic hose, Valves	No Leaks							
2	Compressed Air Pressure	0.6-0.8 MPa							
3	Availability of cooling water	Available							
4	NaClO	> 20 L							
5	NaOH	> 20 L							
6	Plant Steam Pressure (on the header)	3-4 Bar							
7	Press. Diff. of 5 Micron Filter Circulation On	≤ 7 psi							
	Press. Diff. of 5 Micron Filter Idle/Standby	Monitoring Only							
8	UV Lamp Hour Meter	< 8500 h							
9	PW Temp. On the PW Tank	≤ 40 °C							
10	PW Temp. On the looping return 1st floor	Monitoring Only							
11	PW Temp. On the looping return 2nd floor	Monitoring Only							
12	PW Conductivity after EDI Circulation On	< 1.2 µ/cm							
	PW Conductivity after EDI Idle/Stanby	Monitoring Only							
13	PW Cond. On the looping return 1st floor	< 1.2 µ/cm							
14	PW Cond. On the looping return 2nd floor	< 1.2 µ/cm							
15	PW Loop 1 Flow Velocity	> 1 m/s							
16	PW Loop 2 Flow Velocity	> 1 m/s							
17	pH	7-9							
18	Soft Water total hardnees (once a day)	< 10 ppm							
19	Differential Pressure MMF	< 0.2 MPa							
20	Differential Pressure ACF	< 0.2 MPa							
21	Cleanliness of machine and room	Clean							

Notes : PW Conductivity dan PW Conductivity action limit : Sesuai Lampiran SOP/EBI/EN-015-L07 UV Lamp hour meter alert limit : 8000 h, UV Lamp hour meter action limit : 8500 h PW Loop Flow Velocity alert limit : 1.1 m/s, PW Loop Flow Velocity action limit : 1 m/s Pressure Diff. of 5 micron filter alert limit : 6 psi Pressure Diff. of 5 micron filter action limit : 7 psi Pembuatan NaOH dan NaClO mengacu pada SOP/EBI/EN-015. point 7.1.1.2 dan 7.1.1.3.	Operator : Shift 1 Shift 2 Shift 3 () () ()	Checked By : ()
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 PT ETANA BIOTECHNOLOGIES INDONESIA	LOGBOOK / LOGBOOK						Document No. : SOP/EBI/EN-015-F01H
	PEMANTAUAN HARIAN TERHADAP GENERATOR AIR MURNI DAN DISTRIBUSI SKIDNYA DAILY MONITORING OF PW GENERATOR AND THE DISTRIBUTION SKID						

Date :

No	Checklist Items	Standard Value	Time : (Shift 1, Shift 2 dan Shift 3)						Remark
			07:30	11:30	15:30	19:30	23:30	03:30	
1	Piping, Pneumatic hose, Valves	No Leaks							
2	Compressed Air Pressure	0.6-0.8 MPa							
3	Availability of cooling water	Available							
4	NaClO	> 20 L							
5	NaOH	> 20 L							
6	Plant Steam Pressure (on the header)	3-4 Bar							
7	Press. Diff. of 5 Micron Filter Circulation On	≤ 7 psi							
	Press. Diff. of 5 Micron Filter Idle/Standby	Monitoring Only							
8	UV Lamp Hour Meter	< 8500 h							
9	PW Temp. On the PW Tank	≤ 40 °C							
10	PW Temp. On the looping return 1st floor	Monitoring Only							
11	PW Temp. On the looping return 2nd floor	Monitoring Only							
12	PW Conductivity after EDI Circulation On	< 1.2 µ/cm							
	PW Conductivity after EDI Idle/Stanby	Monitoring Only							
13	PW Cond. On the looping return 1st floor	< 1.2 µ/cm							
14	PW Cond. On the looping return 2nd floor	< 1.2 µ/cm							
15	PW Loop 1 Flow Velocity	> 1 m/s							
16	PW Loop 2 Flow Velocity	> 1 m/s							
17	pH	7-9							
18	Soft Water total hardnees (once a day)	< 10 ppm							
19	Differential Pressure MMF	< 0.2 MPa							
20	Differential Pressure ACF	< 0.2 MPa							
21	Cleanliness of machine and room	Clean							

Notes : PW Conductivity dan PW Conductivity action limit : Sesuai Lampiran SOP/EBI/EN-015-L07 UV Lamp hour meter alert limit : 8000 h, UV Lamp hour meter action limit : 8500 h PW Loop Flow Velocity alert limit : 1.1 m/s, PW Loop Flow Velocity action limit : 1 m/s Pressure Diff. of 5 micron filter alert limit : 6 psi Pressure Diff. of 5 micron filter action limit : 7 psi Pembuatan NaOH dan NaClO mengacu pada SOP/EBI/EN-015. point 7.1.1.2 dan 7.1.1.3.	Operator : Shift 1 Shift 2 Shift 3 () () ()	Checked By : ()
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 PT ETANA BIOTECHNOLOGIES INDONESIA	LOGBOOK / LOGBOOK						Document No. : SOP/EBI/EN-015-F01H
	PEMANTAUAN HARIAN TERHADAP GENERATOR AIR MURNI DAN DISTRIBUSI SKIDNYA DAILY MONITORING OF PW GENERATOR AND THE DISTRIBUTION SKID						

Date :

No	Checklist Items	Standard Value	Time : (Shift 1, Shift 2 dan Shift 3)						Remark
			07:30	11:30	15:30	19:30	23:30	03:30	
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7	Press. Diff. of 5 Micron Filter Circulation On	≤ 7 psi							
	Press. Diff. of 5 Micron Filter Idle/Standby	Monitoring Only							
8	UV Lamp Hour Meter	< 8500 h							
9	PW Temp. On the PW Tank	≤ 40 °C							
10	PW Temp. On the looping return 1st floor	Monitoring Only							
11	PW Temp. On the looping return 2nd floor	Monitoring Only							
12	PW Conductivity after EDI Circulation On	< 1.2 µ/cm							
	PW Conductivity after EDI Idle/Stanby	Monitoring Only							
13	PW Cond. On the looping return 1st floor	< 1.2 µ/cm							
14	PW Cond. On the looping return 2nd floor	< 1.2 µ/cm							
15	PW Loop 1 Flow Velocity	> 1 m/s							
16	PW Loop 2 Flow Velocity	> 1 m/s							
17	pH	7-9							
18	Soft Water total hardnees (once a day)	< 10 ppm							
19	Differential Pressure MMF	< 0.2 MPa							
20	Differential Pressure ACF	< 0.2 MPa							
21	Cleanliness of machine and room	Clean							
Notes : PW Conductivity dan PW Conductivity action limit : Sesuai Lampiran SOP/EBI/EN-015-L07 UV Lamp hour meter alert limit : 8000 h, UV Lamp hour meter action limit : 8500 h PW Loop Flow Velocity alert limit : 1.1 m/s, PW Loop Flow Velocity action limit : 1 m/s Pressure Diff. of 5 micron filter alert limit : 6 psi Pressure Diff. of 5 micron filter action limit : 7 psi Pembuatan NaOH dan NaClO mengacu pada SOP/EBI/EN-015. point 7.1.1.2 dan 7.1.1.3.			Operator : Shift 1 Shift 2 Shift 3 () () ()						Checked By : ()

 PT ETANA BIOTECHNOLOGIES INDONESIA	LOGBOOK / LOGBOOK		Document No. : SOP/EBI/EN-015-F01H
	PEMANTAUAN HARIAN TERHADAP GENERATOR AIR MURNI DAN DISTRIBUSI SKIDNYA DAILY MONITORING OF PW GENERATOR AND THE DISTRIBUTION SKID		

Date :

No	Checklist Items	Standard Value	Time : (Shift 1, Shift 2 dan Shift 3)						Remark
			07:30	11:30	15:30	19:30	23:30	03:30	
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	Press. Diff. of 5 Micron Filter Idle/Standby	Monitoring Only							
8	UV Lamp Hour Meter	< 8500 h							
9	PW Temp. On the PW Tank	≤ 40 °C							
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12	PW Conductivity after EDI Circulation On	< 1.2 µ/cm							
	PW Conductivity after EDI Idle/Stanby	Monitoring Only							
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15	PW Loop 1 Flow Velocity	> 1 m/s							
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17	pH	7-9							
18	Soft Water total hardnees (once a day)	< 10 ppm							
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 PT ETANA BIOTECHNOLOGIES INDONESIA	LOGBOOK / LOGBOOK						Document No. : SOP/EBI/EN-015-F01H
	PEMANTAUAN HARIAN TERHADAP GENERATOR AIR MURNI DAN DISTRIBUSI SKIDNYA DAILY MONITORING OF PW GENERATOR AND THE DISTRIBUTION SKID						

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	Press. Diff. of 5 Micron Filter Idle/Standby	Monitoring Only							
8	UV Lamp Hour Meter	< 8500 h							
9	PW Temp. On the PW Tank	≤ 40 °C							
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12	PW Conductivity after EDI Circulation On	< 1.2 µ/cm							
	PW Conductivity after EDI Idle/Stanby	Monitoring Only							
13	PW Cond. On the looping return 1st floor	< 1.2 µ/cm							
14	PW Cond. On the looping return 2nd floor	< 1.2 µ/cm							
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17	pH	7-9							
18	Soft Water total hardnees (once a day)	< 10 ppm							
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 PT ETANA BIOTECHNOLOGIES INDONESIA	LOGBOOK / LOGBOOK						Document No. : SOP/EBI/EN-015-F01H
	PEMANTAUAN HARIAN TERHADAP GENERATOR AIR MURNI DAN DISTRIBUSI SKIDNYA DAILY MONITORING OF PW GENERATOR AND THE DISTRIBUTION SKID						

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13	PW Cond. On the looping return 1st floor	< 1.2 µ/cm							
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15	PW Loop 1 Flow Velocity	> 1 m/s							
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17	pH	7-9							
18	Soft Water total hardnees (once a day)	< 10 ppm							
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21	Cleanliness of machine and room	Clean							

Notes : PW Conductivity dan PW Conductivity action limit : Sesuai Lampiran SOP/EBI/EN-015-L07 UV Lamp hour meter alert limit : 8000 h, UV Lamp hour meter action limit : 8500 h PW Loop Flow Velocity alert limit : 1.1 m/s, PW Loop Flow Velocity action limit : 1 m/s Pressure Diff. of 5 micron filter alert limit : 6 psi Pressure Diff. of 5 micron filter action limit : 7 psi Pembuatan NaOH dan NaClO mengacu pada SOP/EBI/EN-015. point 7.1.1.2 dan 7.1.1.3.	Operator : Shift 1 Shift 2 Shift 3 () () ()	Checked By : ()
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 PT ETANA BIOTECHNOLOGIES INDONESIA	LOGBOOK / LOGBOOK						Document No. : SOP/EBI/EN-015-F01H
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8	UV Lamp Hour Meter	< 8500 h							
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12	PW Conductivity after EDI Circulation On	< 1.2 μ/cm							
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13	PW Cond. On the looping return 1st floor	< 1.2 μ/cm							
14	PW Cond. On the looping return 2nd floor	< 1.2 μ/cm							
15	PW Loop 1 Flow Velocity	> 1 m/s							
16	PW Loop 2 Flow Velocity	> 1 m/s							
17	pH	7-9							
18	Soft Water total hardnees (once a day)	< 10 ppm							
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 PT ETANA BIOTECHNOLOGIES INDONESIA	LOGBOOK / LOGBOOK						Document No. : SOP/EBI/EN-015-F01H
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15	PW Loop 1 Flow Velocity	> 1 m/s							
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	PEMANTAUAN HARIAN TERHADAP GENERATOR AIR MURNI DAN DISTRIBUSI SKIDNYA DAILY MONITORING OF PW GENERATOR AND THE DISTRIBUTION SKID						

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21	Cleanliness of machine and room	Clean							

Notes :

PW Conductivity dan PW Conductivity action limit : Sesuai Lampiran SOP/EBI/EN-015-L07
 UV Lamp hour meter alert limit : 8000 h, UV Lamp hour meter action limit : 8500 h
 PW Loop Flow Velocity alert limit : 1.1 m/s, PW Loop Flow Velocity action limit : 1 m/s
 Pressure Diff. of 5 micron filter alert limit : 6 psi
 Pressure Diff. of 5 micron filter action limit : 7 psi
 Pembuatan NaOH dan NaClO mengacu pada SOP/EBI/EN-015. point 7.1.1.2 dan 7.1.1.3.

Operator :

Shift 1

Shift 2

Shift 3

Checked By :

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 PT ETANA BIOTECHNOLOGIES INDONESIA	LOGBOOK / LOGBOOK						Document No. : SOP/EBI/EN-015-F01H
	PEMANTAUAN HARIAN TERHADAP GENERATOR AIR MURNI DAN DISTRIBUSI SKIDNYA DAILY MONITORING OF PW GENERATOR AND THE DISTRIBUTION SKID						

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11	PW Temp. On the looping return 2nd floor	Monitoring Only							
12	PW Conductivity after EDI Circulation On	< 1.2 μ/cm							
	PW Conductivity after EDI Idle/Stanby	Monitoring Only							
13	PW Cond. On the looping return 1st floor	< 1.2 μ/cm							
14	PW Cond. On the looping return 2nd floor	< 1.2 μ/cm							
15	PW Loop 1 Flow Velocity	> 1 m/s							
16	PW Loop 2 Flow Velocity	> 1 m/s							
17	pH	7-9							
18	Soft Water total hardnees (once a day)	< 10 ppm							
19	Differential Pressure MMF	< 0.2 MPa							
20	Differential Pressure ACF	< 0.2 MPa							
21	Cleanliness of machine and room	Clean							

Notes :

PW Conductivity dan PW Conductivity action limit : Sesuai Lampiran SOP/EBI/EN-015-L07

UV Lamp hour meter alert limit : 8000 h, UV Lamp hour meter action limit : 8500 h

PW Loop Flow Velocity alert limit : 1.1 m/s, PW Loop Flow Velocity action limit : 1 m/s

Pressure Diff. of 5 micron filter alert limit : 6 psi

Pressure Diff. of 5 micron filter action limit : 7 psi

Pembuatan NaOH dan NaClO mengacu pada SOP/EBI/EN-015. point 7.1.1.2 dan 7.1.1.3.

Operator :

Shift 1	Shift 2	Shift 3
()	()	()

Checked By :

()

 PT ETANA BIOTECHNOLOGIES INDONESIA	LOGBOOK / LOGBOOK						Document No. : SOP/EBI/EN-015-F01H
	PEMANTAUAN HARIAN TERHADAP GENERATOR AIR MURNI DAN DISTRIBUSI SKIDNYA DAILY MONITORING OF PW GENERATOR AND THE DISTRIBUTION SKID						

Date :

No	Checklist Items	Standard Value	Time : (Shift 1, Shift 2 dan Shift 3)						Remark
			07:30	11:30	15:30	19:30	23:30	03:30	
1	Piping, Pneumatic hose, Valves	No Leaks							
2	Compressed Air Pressure	0.6-0.8 MPa							
3	Availability of cooling water	Available							
4	NaClO	> 20 L							
5	NaOH	> 20 L							
6	Plant Steam Pressure (on the header)	3-4 Bar							
7	Press. Diff. of 5 Micron Filter Circulation On	≤ 7 psi							
	Press. Diff. of 5 Micron Filter Idle/Standby	Monitoring Only							
8	UV Lamp Hour Meter	< 8500 h							
9	PW Temp. On the PW Tank	≤ 40 °C							
10	PW Temp. On the looping return 1st floor	Monitoring Only							
11	PW Temp. On the looping return 2nd floor	Monitoring Only							
12	PW Conductivity after EDI Circulation On	< 1.2 µ/cm							
	PW Conductivity after EDI Idle/Stanby	Monitoring Only							
13	PW Cond. On the looping return 1st floor	< 1.2 µ/cm							
14	PW Cond. On the looping return 2nd floor	< 1.2 µ/cm							
15	PW Loop 1 Flow Velocity	> 1 m/s							
16	PW Loop 2 Flow Velocity	> 1 m/s							
17	pH	7-9							
18	Soft Water total hardnees (once a day)	< 10 ppm							
19	Differential Pressure MMF	< 0.2 MPa							
20	Differential Pressure ACF	< 0.2 MPa							
21	Cleanliness of machine and room	Clean							
Notes : PW Conductivity dan PW Conductivity action limit : Sesuai Lampiran SOP/EBI/EN-015-L07 UV Lamp hour meter alert limit : 8000 h, UV Lamp hour meter action limit : 8500 h PW Loop Flow Velocity alert limit : 1.1 m/s, PW Loop Flow Velocity action limit : 1 m/s Pressure Diff. of 5 micron filter alert limit : 6 psi Pressure Diff. of 5 micron filter action limit : 7 psi Pembuatan NaOH dan NaClO mengacu pada SOP/EBI/EN-015. point 7.1.1.2 dan 7.1.1.3.			Operator : Shift 1 Shift 2 Shift 3 () () ()						Checked By : ()

 PT ETANA BIOTECHNOLOGIES INDONESIA	LOGBOOK / LOGBOOK						Document No. : SOP/EBI/EN-015-F01H
	PEMANTAUAN HARIAN TERHADAP GENERATOR AIR MURNI DAN DISTRIBUSI SKIDNYA DAILY MONITORING OF PW GENERATOR AND THE DISTRIBUTION SKID						

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			07:30	11:30	15:30	19:30	23:30	03:30	
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2	Compressed Air Pressure	0.6-0.8 MPa							
3	Availability of cooling water	Available							
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5	NaOH	> 20 L							
6	Plant Steam Pressure (on the header)	3-4 Bar							
7	Press. Diff. of 5 Micron Filter Circulation On	≤ 7 psi							
	Press. Diff. of 5 Micron Filter Idle/Standby	Monitoring Only							
8	UV Lamp Hour Meter	< 8500 h							
9	PW Temp. On the PW Tank	≤ 40 °C							
10	PW Temp. On the looping return 1st floor	Monitoring Only							
11	PW Temp. On the looping return 2nd floor	Monitoring Only							
12	PW Conductivity after EDI Circulation On	< 1.2 µ/cm							
	PW Conductivity after EDI Idle/Stanby	Monitoring Only							
13	PW Cond. On the looping return 1st floor	< 1.2 µ/cm							
14	PW Cond. On the looping return 2nd floor	< 1.2 µ/cm							
15	PW Loop 1 Flow Velocity	> 1 m/s							
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17	pH	7-9							
18	Soft Water total hardnees (once a day)	< 10 ppm							
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20	Differential Pressure ACF	< 0.2 MPa							
21	Cleanliness of machine and room	Clean							
Notes : PW Conductivity dan PW Conductivity action limit : Sesuai Lampiran SOP/EBI/EN-015-L07 UV Lamp hour meter alert limit : 8000 h, UV Lamp hour meter action limit : 8500 h PW Loop Flow Velocity alert limit : 1.1 m/s, PW Loop Flow Velocity action limit : 1 m/s Pressure Diff. of 5 micron filter alert limit : 6 psi Pressure Diff. of 5 micron filter action limit : 7 psi Pembuatan NaOH dan NaClO mengacu pada SOP/EBI/EN-015. point 7.1.1.2 dan 7.1.1.3.			Operator : Shift 1 Shift 2 Shift 3 () () ()						Checked By : ()

 PT ETANA BIOTECHNOLOGIES INDONESIA	LOGBOOK / LOGBOOK						Document No. : SOP/EBI/EN-015-F01H
	PEMANTAUAN HARIAN TERHADAP GENERATOR AIR MURNI DAN DISTRIBUSI SKIDNYA DAILY MONITORING OF PW GENERATOR AND THE DISTRIBUTION SKID						

Date :

No	Checklist Items	Standard Value	Time : (Shift 1, Shift 2 dan Shift 3)						Remark
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2	Compressed Air Pressure	0.6-0.8 MPa							
3	Availability of cooling water	Available							
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6	Plant Steam Pressure (on the header)	3-4 Bar							
7	Press. Diff. of 5 Micron Filter Circulation On	≤ 7 psi							
	Press. Diff. of 5 Micron Filter Idle/Standby	Monitoring Only							
8	UV Lamp Hour Meter	< 8500 h							
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15	PW Loop 1 Flow Velocity	> 1 m/s							
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17	pH	7-9							
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21	Cleanliness of machine and room	Clean							

Notes : PW Conductivity dan PW Conductivity action limit : Sesuai Lampiran SOP/EBI/EN-015-L07 UV Lamp hour meter alert limit : 8000 h, UV Lamp hour meter action limit : 8500 h PW Loop Flow Velocity alert limit : 1.1 m/s, PW Loop Flow Velocity action limit : 1 m/s Pressure Diff. of 5 micron filter alert limit : 6 psi Pressure Diff. of 5 micron filter action limit : 7 psi Pembuatan NaOH dan NaClO mengacu pada SOP/EBI/EN-015. point 7.1.1.2 dan 7.1.1.3.	Operator : Shift 1 Shift 2 Shift 3 () () ()	Checked By : ()
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 PT ETANA BIOTECHNOLOGIES INDONESIA	LOGBOOK / LOGBOOK						Document No. : SOP/EBI/EN-015-F01H
	PEMANTAUAN HARIAN TERHADAP GENERATOR AIR MURNI DAN DISTRIBUSI SKIDNYA DAILY MONITORING OF PW GENERATOR AND THE DISTRIBUTION SKID						

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15	PW Loop 1 Flow Velocity	> 1 m/s							
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 PT ETANA BIOTECHNOLOGIES INDONESIA	LOGBOOK / LOGBOOK		Document No. : SOP/EBI/EN-015-F01H
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